

07 Executive Summary (FRA DSS)

Disclosure: all results use the synthetic fra_synthetic_claims dataset. Scores may reflect generator rules. Real-world performance is unknown. This demo must not drive real claim decisions.

1. Problem

FRA Atlas DSS predicts claim outcome status (Approved, Pending, Rejected) to support officer triage for Ministry of Tribal Affairs workflows (SIH12508).

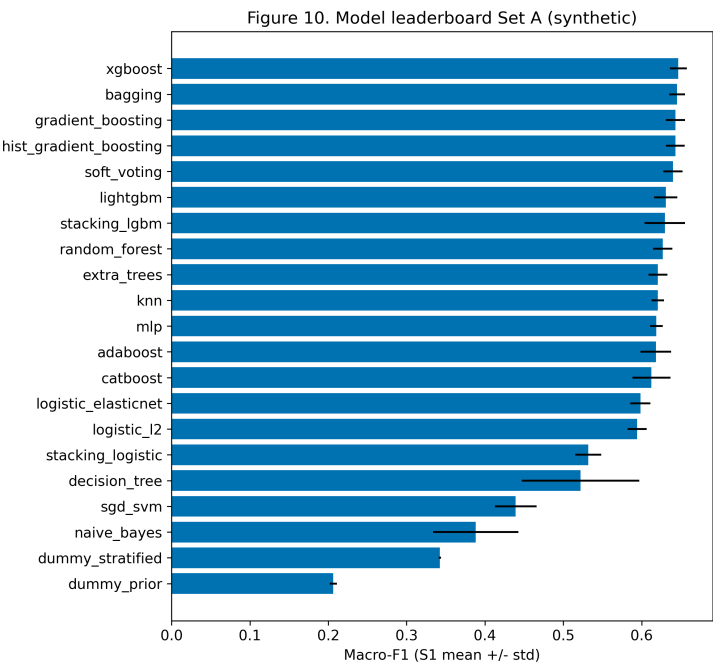
2. Approach

Leakage-safe Set A (filing-time) features, grouped district CV, a model zoo spanning baselines, bagging, boosting, voting and stacking, plus a LightGBM champion shipped for TreeExplainer compatibility.

3. Headline results

Existing baseline macro-F1: 0.5690448456514766. Champion (S1 Set A): xgboost with S1 macro-F1 0.6468642536953271 and final-test macro-F1 0.5690448456514766.

Set B minus Set A gap: 0.18053581159110665. Positive gap means process-stage features lift macro-F1. That lift is partly circular because committee level and processing days are correlated with outcome.



4. DSS and limits

The DSS returns calibrated-style probabilities, SHAP drivers, scheme convergence hints, ETA on resolved claims, and optional what-if advice on actionable fields only.

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